Cardiovascular System (CARD) · NPLEX-P1-CARD-13

CV-13 Cardiovascular Biochemistry and Nutrition

Brain dumps can be right in more than one way. Grade the ideas and the relationships, not the wording. A response is competent when it carries the essentials below.

NPLEX-P1-CARD-13-01-C01-BD1

Brain dump 1 · foundational retrieval: Build the lipoprotein table from memory

A competent response contains

· Chylomicron: made by enterocytes, carries apo B-48 with C-II and E borrowed from HDL, delivers dietary triglyceride to muscle and adipose tissue, and its remnant returns to the liver.

· VLDL: made by the liver, carries apo B-100 with C-II and E from HDL, and carries liver-made triglyceride to tissues.

· IDL: formed from VLDL in plasma, carries apo B-100 and E, and is either taken up by the liver, about half of it, or converted to LDL.

· LDL: formed from IDL, carries apo B-100 only, and is the main plasma carrier of cholesterol to the liver and peripheral cells.

· HDL: made by liver and intestine as lipid-poor apo A-I, carries A-I, and runs reverse cholesterol transport while lending C-II and E to other particles.

· As lipoprotein lipase removes triglyceride, particles become smaller, denser, and proportionally richer in cholesterol, so density rises from chylomicrons through VLDL, IDL, and LDL to HDL, which has the most protein relative to lipid.

· Apo B-48 is needed for chylomicron assembly and lacks the LDL receptor-binding region; apo B-100 is the LDL receptor ligand; apo C-II activates lipoprotein lipase; apo E mediates hepatic uptake of remnants and IDL; apo A-I activates LCAT and accepts cholesterol through ABCA1.

· LCAT is a plasma enzyme that esterifies free cholesterol on the HDL surface so it moves into the core; ACAT esterifies cholesterol inside cells for storage.

· Apo B never leaves its particle, so each chylomicron carries one B-48 and each VLDL, IDL, and LDL one B-100.

Distinctions that matter

· Apo B-48 and apo B-100 come from the same APOB gene; intestinal RNA editing changes one cytosine to uracil and creates a stop codon at about 48% of full length.

· LCAT works outside the cell on HDL and ACAT works inside the cell, so the two esterification steps serve opposite purposes: keeping the efflux gradient open against storing excess.

Common misconceptions

· Writing that HDL is dense because it is fat-rich. It is the most protein-rich and lipid-poor particle, which is exactly why it is the densest.

· Treating apo C-II and apo E as native to chylomicrons and VLDL. HDL lends them, which is why HDL loss slows triglyceride clearance as well as reverse transport.

Where it is taught: Section C01, opening paragraph and the lipoprotein table of the CV-13 Cardiovascular Biochemistry and Nutrition notes.

NPLEX-P1-CARD-13-01-C01-BD2

Brain dump 2 · integration and application: Three broken steps in lipid transport

A competent response contains

· Patient 1 without apo C-II: lipoprotein lipase cannot be activated, so chylomicron and VLDL triglyceride is not hydrolyzed and chylomicrons accumulate; triglyceride rises most, and refrigerated plasma shows a creamy top layer because chylomicrons are the largest and lightest particle.

· Patient 2 without ABCA1: lipid-poor apo A-I cannot collect free cholesterol and phospholipid from cells, so HDL is nearly absent, which is Tangier disease; cholesterol carried on HDL is very low and plasma shows neither a cream layer nor marked turbidity.

· Patient 3 with apo E2 homozygosity: remnants and IDL bind hepatic receptors poorly and accumulate, raising cholesterol and triglyceride to similar levels, and plasma is turbid rather than creamy because the particles are remnants rather than intact chylomicrons.

· Particles upstream cannot move on because each step needs the one before it: no activator means no hydrolysis, no acceptor means no efflux, and no receptor ligand means no uptake.

· High triglyceride with low HDL arises because CETP transfers cholesteryl ester from HDL to abundant apo B particles in exchange for triglyceride, and hepatic lipase then removes that triglyceride, leaving small HDL that is cleared quickly.

· Normal plasma is free of chylomicrons after a 12-hour fast, so finding them in a fasting sample means clearance has failed.

Distinctions that matter

· Chylomicrons float and VLDL stays suspended, so the appearance of refrigerated plasma separates a clearance defect in the exogenous pathway from overproduction in the endogenous one.

· Apo E2 homozygosity usually needs a second metabolic load such as obesity, diabetes, hypothyroidism, or alcohol before lipids rise.

Common misconceptions

· Expecting severe atherosclerosis whenever triglyceride is very high. Chylomicrons are too large to enter the intima efficiently, so type I disease adds little atherosclerotic risk.

· Treating a low HDL as an independent second disorder when triglyceride is high. The CETP and hepatic lipase sequence links the two.

Where it is taught: Section C01, HDL and reverse cholesterol transport paragraph, with the Fredrickson table in C02 of the CV-13 Cardiovascular Biochemistry and Nutrition notes.

NPLEX-P1-CARD-13-01-C01-RT01

Retrieval exercise: Rebuild the exogenous lipoprotein pathway

A competent response contains

· Step 1: enterocytes resynthesize absorbed fat into triglyceride and microsomal triglyceride transfer protein loads it, with cholesteryl ester and fat-soluble vitamins, onto apo B-48 as a chylomicron.

· Step 2: the particle travels through lacteals and the thoracic duct, because chylomicrons are too large to enter blood capillaries, which is why dietary fat reaches peripheral tissues before the liver.

· Step 3: HDL donates apo C-II and apo E.

· Step 4: apo C-II activates lipoprotein lipase on the luminal surface of capillary endothelium, which hydrolyzes core triglyceride and releases fatty acids.

· Step 5: those fatty acids enter adipose tissue, skeletal muscle, and heart.

· Step 6: the liver takes up the remnant through apo E.

· Lipoprotein lipase deficiency stops the pathway at step 4, so chylomicrons accumulate; normal plasma is clear of them after a 12-hour fast.

· Endogenous pathway: the liver secretes VLDL on apo B-100, lipoprotein lipase converts VLDL to IDL, about half of IDL is taken up through apo E, hepatic lipase converts the rest to LDL, and LDL loses apo E and is cleared through apo B-100 binding the LDL receptor, mostly in the liver, with adrenal cortex and gonads taking some for steroid synthesis.

· Insulin raises adipose lipoprotein lipase after a meal, while the cardiac enzyme has a lower Km and keeps supplying the heart during fasting.

Distinctions that matter

· Lipoprotein lipase acts on capillary endothelium and makes IDL from VLDL; hepatic lipase acts at the liver and makes LDL from IDL.

· LDL lasts days in plasma while chylomicrons are cleared within hours, which is why fasting lipid studies measure LDL reliably.

Common misconceptions

· Drawing chylomicrons entering the portal vein. They enter lymph and reach the blood through the thoracic duct, bypassing the first pass through the liver.

· Saying LDL is secreted by the liver. The liver secretes VLDL, and LDL is formed in plasma from IDL.

Where it is taught: Section C01, exogenous pathway flow diagram and the paragraph that follows it of the CV-13 Cardiovascular Biochemistry and Nutrition notes.

NPLEX-P1-CARD-13-01-C01-Q01

Board-style question: Reverse cholesterol transport fails at its first step

Correct answer: A. Transfer of free cholesterol and phospholipid from the cell membrane to lipid-poor apolipoprotein A-I

Why it is right: ABCA1 moves free cholesterol and phospholipid from the cell membrane onto lipid-poor apo A-I, so losing it leaves HDL almost absent and traps cholesteryl ester in macrophages and tonsillar tissue, which gives the orange tonsils and the neuropathy of Tangier disease.

B is wrong: Loss of LCAT would also lower HDL cholesterol, but the esterification step comes after efflux and would not fill macrophages with ester or give orange tonsils.

C is wrong: Lipoprotein lipase failure accumulates chylomicrons and raises triglyceride above 1,000 mg/dL with a creamy plasma layer, which this triglyceride of 210 mg/dL does not match.

D is wrong: Failure of apo E mediated remnant uptake raises both cholesterol and triglyceride to similar levels with palmar xanthomas, not an isolated HDL of 3 mg/dL.

A competent response contains

· Lipid-poor apo A-I collects free cholesterol and phospholipid from cells through ABCA1, and ABCA1 is absent in Tangier disease, in which HDL is nearly absent.

· Peripheral cells cannot degrade the sterol ring, so cholesterol that cannot be exported stays stored as cholesteryl ester, which is why macrophages and tonsillar tissue fill with it.

Distinctions that matter

· An efflux defect lowers HDL cholesterol without raising LDL, which separates it from the disorders that accumulate apo B particles.

Common misconceptions

· Reading a very low HDL as a consequence of the raised triglyceride alone. Here the first step of efflux is missing, and the stored ester in macrophages points to it.

Where it is taught: Section C01, HDL and reverse cholesterol transport paragraph of the CV-13 Cardiovascular Biochemistry and Nutrition notes.

NPLEX-P1-CARD-13-01-C02-BD1

Brain dump 1 · foundational retrieval: Build the Fredrickson table from memory

A competent response contains

· Type I familial chylomicronemia, autosomal recessive: lipoprotein lipase or apo C-II deficiency raises chylomicrons with triglyceride often above 1,000 mg/dL, causing pancreatitis, eruptive xanthomas, and lipemia retinalis with little added atherosclerosis.

· Type IIa familial hypercholesterolemia, autosomal dominant: a defective LDL receptor, or abnormal apo B-100, or gain-of-function PCSK9, raises LDL with high cholesterol and normal triglyceride, causing premature atherosclerosis, tendon xanthomas, xanthelasma, and early corneal arcus.

· Type IIb combined: hepatic overproduction of apo B particles, usually polygenic, raises LDL and VLDL with both lipids up and premature atherosclerosis.

· Type III dysbetalipoproteinemia: apo E2 homozygosity plus a second metabolic factor raises remnants and IDL with both lipids raised to similar levels, giving palmar and tuberoeruptive xanthomas and premature coronary and peripheral disease.

· Type IV familial hypertriglyceridemia: hepatic VLDL overproduction raises triglyceride with cholesterol near normal, causing pancreatitis if very high and linked to insulin resistance.

· Type V: VLDL overproduction with impaired clearance raises both VLDL and chylomicrons, with pancreatitis and eruptive xanthomas.

· Desirable adult values: total cholesterol below 200 mg/dL, LDL below 100 mg/dL, HDL at least 40 mg/dL in men and 50 mg/dL in women, fasting triglyceride below 150 mg/dL.

· Secondary causes to exclude first: diabetes, hypothyroidism, nephrotic syndrome, and alcohol.

Distinctions that matter

· High triglyceride points to chylomicrons or VLDL, and high cholesterol with a normal triglyceride points to LDL, so the lipid pattern names the particle before the defect.

· Type I and type IV both raise triglyceride, but one is a clearance failure of chylomicrons and the other an overproduction of VLDL.

Common misconceptions

· Treating the Fredrickson types as single diseases. They describe patterns of raised lipoproteins, and several molecular causes can give the same type.

· Calling a pattern familial before excluding diabetes, hypothyroidism, nephrotic syndrome, and alcohol.

Where it is taught: Section C02, opening paragraph and the Fredrickson table of the CV-13 Cardiovascular Biochemistry and Nutrition notes.

NPLEX-P1-CARD-13-01-C02-BD2

Brain dump 2 · integration and application: Name the type from the numbers

A competent response contains

· Patient 1: type I, chylomicrons accumulating, lipoprotein lipase or apo C-II deficiency, settled by the creamy floating layer over a clear infranatant in a fasting sample; the modest cholesterol rise is the cholesterol each chylomicron carries.

· Patient 2: type IV, VLDL accumulating, hepatic VLDL overproduction, settled by uniform turbidity with triglyceride raised and cholesterol near normal; the high triglyceride, low HDL, small dense LDL pattern is the dyslipidemia of the metabolic syndrome.

· Patient 3: type III, remnants and IDL accumulating, apo E2 homozygosity with a second metabolic load, settled by palmar xanthomas with cholesterol and triglyceride raised to similar levels.

· Patient 4: type IIa, LDL accumulating, a defective LDL receptor, settled by tendon xanthomas with a very high cholesterol, a normal triglyceride, and a first-degree relative with early coronary disease.

· Pancreatitis in patient 1: pancreatic lipase hydrolyzes chylomicron triglyceride in pancreatic capillaries, and the released free fatty acids injure acinar cells.

· Atherosclerotic risk is not greatly raised in patient 1 because chylomicrons are too large to enter the intima efficiently.

· In patient 4 the hepatocytes have too few receptors, so they behave as if cholesterol-poor and keep synthesizing cholesterol and secreting VLDL, which adds still more LDL.

Distinctions that matter

· Which lipid is higher, and the appearance of refrigerated plasma, separate the triglyceride disorders from each other before any genetic test.

· Palmar and tuberoeruptive xanthomas belong to remnant accumulation, while tendon xanthomas and xanthelasma belong to LDL accumulation.

Common misconceptions

· Reading a high plasma cholesterol as a signal that will switch off hepatic synthesis. The hepatocyte senses its own sterol, not the plasma level, so receptor loss leaves synthesis running.

· Assuming any very high triglyceride causes atherosclerosis. Particle size decides whether the lipid can be retained in the intima.

Where it is taught: Section C02, paragraph on mechanisms and the worked example on classifying a dyslipidemia of the CV-13 Cardiovascular Biochemistry and Nutrition notes.

NPLEX-P1-CARD-13-01-C02-CP01

Comparison table: Type I against type IIa against type III

A competent response contains

· Defect and inheritance: type I is lipoprotein lipase or apo C-II deficiency, autosomal recessive; type IIa is a defective LDL receptor, or abnormal apo B-100, or gain-of-function PCSK9, autosomal dominant; type III is apo E2 homozygosity plus a second metabolic factor.

· Particle: type I accumulates chylomicrons, type IIa accumulates LDL, type III accumulates remnants and IDL.

· Lipid pattern: type I has triglyceride often above 1,000 mg/dL with a modest cholesterol rise; type IIa has high cholesterol with a normal triglyceride; type III has cholesterol and triglyceride raised to similar levels.

· Refrigerated plasma: type I shows a creamy floating layer over clear plasma; type IIa leaves plasma clear; type III leaves plasma turbid from remnant particles.

· Skin and eye findings: type I gives eruptive xanthomas and lipemia retinalis; type IIa gives tendon xanthomas on the Achilles and hand extensor tendons, xanthelasma, and early corneal arcus; type III gives palmar and tuberoeruptive xanthomas.

· Atherosclerotic risk: little added risk in type I because chylomicrons are too large to enter the intima efficiently; premature atherosclerosis in type IIa because LDL is retained in the intima; premature coronary and peripheral disease in type III because remnants are also retained.

· What else must be present: type I and type IIa express with the defect alone, while most apo E2 homozygotes have near-normal lipids until obesity, diabetes, hypothyroidism, or alcohol adds a second load.

Distinctions that matter

· Particle size decides atherosclerotic risk, which is why the highest triglyceride of the three carries the lowest arterial risk.

· Type III needs a second metabolic factor to appear, so a normal lipid panel does not exclude the genotype.

Common misconceptions

· Ranking risk by triglyceride level. The retained particle, not the lipid number, sets arterial risk.

· Expecting xanthomas to look the same across the types. Their site is the clue: tendons in type IIa, palmar creases in type III, eruptive skin lesions in type I.

Where it is taught: Section C02, Fredrickson table, mechanism paragraph, and worked example of the CV-13 Cardiovascular Biochemistry and Nutrition notes.

NPLEX-P1-CARD-13-01-C03-BD1

Brain dump 1 · foundational retrieval: Cholesterol synthesis and the four controls on the reductase

A competent response contains

· Three cytosolic acetyl-CoA form HMG-CoA through cytosolic HMG-CoA synthase; the mitochondrial isoform of that synthase instead makes HMG-CoA for ketogenesis.

· HMG-CoA reductase, in the endoplasmic reticulum membrane, reduces HMG-CoA to mevalonate using 2 NADPH, and this is the rate-limiting and main regulated step.

· Order after that: mevalonate, isoprene units, farnesyl pyrophosphate, squalene, lanosterol, cholesterol.

· Farnesyl pyrophosphate is the branch point, also supplying ubiquinone, dolichol, and protein lipid anchors, so blocking the reductase lowers these as well.

· Control 1, transcription: SREBP-2 raises reductase transcription when cell sterol is low.

· Control 2, degradation: accumulating sterols make the enzyme bind Insig, which leads to ubiquitination and proteasomal degradation.

· Control 3, phosphorylation: AMPK phosphorylates and inactivates the enzyme when energy is short.

· Control 4, hormones: insulin and thyroid hormone increase activity or expression, while glucagon and glucocorticoids decrease them.

Distinctions that matter

· Sterols act at two different levels, raising degradation of existing enzyme and lowering transcription of new enzyme, which is why the response is both fast and sustained.

· AMPK reports energy status while sterols report cholesterol status, so the enzyme can be switched off for either reason.

Common misconceptions

· Naming HMG-CoA synthase as the regulated step. The reductase that makes mevalonate is rate-limiting.

· Treating the pathway as making only cholesterol. Ubiquinone, dolichol, and protein lipid anchors branch off at farnesyl pyrophosphate.

Where it is taught: Section C03, synthesis and control of HMG-CoA reductase of the CV-13 Cardiovascular Biochemistry and Nutrition notes.

NPLEX-P1-CARD-13-01-C03-BD2

Brain dump 2 · integration and application: Follow one LDL particle into the cell and out again

A competent response contains

· Apo B-100 binds the LDL receptor, which also binds apo E.

· The receptor's NPXY tail motif gathers it into clathrin-coated pits, and dynamin pinches the vesicle off.

· In the acidified endosome LDL is released from the receptor; the receptor recycles to the surface and LDL goes to the lysosome.

· In the lysosome apo B-100 is degraded and acid lipase frees cholesterol, which leaves through NPC1 and NPC2.

· Blocked steps named in the notes: a defective receptor or defective apo B-100 in familial hypercholesterolemia, gain-of-function PCSK9 which diverts receptors to the lysosome and shortens their life, and NPC1 or NPC2 failure which traps cholesterol in Niemann-Pick type C.

· The released cholesterol suppresses SREBP processing, speeds reductase degradation, and activates ACAT to store the excess as cholesteryl ester.

· Hypothyroidism lowers LDL receptor expression, so clearance falls and plasma LDL rises, which makes it a common secondary cause of high LDL.

· Macrophage scavenger receptors such as SR-A and CD36 take up oxidized LDL and are not downregulated by cell cholesterol, so the cell keeps loading until it is full of cholesteryl ester and becomes a foam cell.

Distinctions that matter

· The receptor recycles and the ligand is degraded, so one receptor can clear many particles unless PCSK9 sends it to the lysosome.

· The LDL receptor is regulated by cell sterol and the scavenger receptor is not, which is the whole reason foam cells form.

Common misconceptions

· Saying the receptor is degraded with its ligand. It releases LDL in the acidic endosome and returns to the surface.

· Treating free cholesterol as the storage form. ACAT esterifies it, because excess free cholesterol is toxic to membranes.

Where it is taught: Section C03, LDL receptor-mediated endocytosis and the paragraph on feedback of the CV-13 Cardiovascular Biochemistry and Nutrition notes.

NPLEX-P1-CARD-13-01-C03-RT01

Retrieval exercise: Rebuild SREBP activation

A competent response contains

· Step 1: endoplasmic reticulum membrane cholesterol falls.

· Step 2: SCAP releases Insig and escorts SREBP-2 to the Golgi.

· Step 3: site-1 protease then site-2 protease cleave SREBP-2.

· Step 4: the freed N-terminal transcription factor enters the nucleus.

· Step 5: it binds sterol regulatory elements.

· Step 6: transcription of the LDL receptor and HMG-CoA reductase rises.

· Step 7: more LDL uptake and more synthesis restore cell sterol.

· When sterol is high, cholesterol-bound SCAP binds Insig and the whole complex stays in the endoplasmic reticulum, so both genes are turned down.

· SREBP-1c mainly controls fatty acid synthesis genes and responds to insulin.

· This is why inhibiting hepatic HMG-CoA reductase lowers plasma LDL: less cell sterol releases SREBP-2, LDL receptor number rises, and clearance increases, which requires receptor genes able to respond.

Distinctions that matter

· SCAP is the cholesterol sensor and escort, Insig is the anchor that holds the complex in the endoplasmic reticulum, and the two proteases do the cutting in a fixed order.

· SREBP-2 serves cholesterol genes and SREBP-1c serves fatty acid genes, so insulin and sterol signals act through different family members.

Common misconceptions

· Describing SREBP as a free cytosolic transcription factor. It starts as a membrane-bound protein and only its cleaved N-terminal fragment reaches the nucleus.

· Expecting the LDL receptor gene and the reductase gene to move in opposite directions. The same signal raises both.

Where it is taught: Section C03, SREBP activation flow diagram and its caption of the CV-13 Cardiovascular Biochemistry and Nutrition notes.

NPLEX-P1-CARD-13-01-C03-Q02

Board-style question: Why one child responds poorly

Correct answer: C. Most of the plasma effect of the drug comes from SREBP-2 driven production of new LDL receptors, which her cells cannot make

Why it is right: Inhibiting the reductase lowers hepatocyte sterol, which releases SREBP-2 and raises LDL receptor transcription, and that extra receptor number is what clears LDL from plasma; a child with two null alleles cannot make the receptor, so only the reduced VLDL output is left.

A is wrong: Hepatic synthesis is not suppressed in this child: the liver takes up little LDL cholesterol, so SREBP-2 stays active and synthesis runs high.

B is wrong: Her apolipoprotein B-100 is normal; the defect is in the receptor gene, and an abnormal apo B-100 is a separate cause of the same lipid pattern.

D is wrong: The drug acts before the farnesyl pyrophosphate branch point, so it lowers ubiquinone as well as sterols rather than acting only on the non-sterol branch.

A competent response contains

· Most of the plasma LDL lowering produced by an HMG-CoA reductase inhibitor comes from lower hepatic sterol releasing SREBP-2, which raises LDL receptor number and increases clearance.

· With two null receptor alleles there is no functional receptor gene to turn on, so only the fall in VLDL output remains, and hepatic synthesis stays high because the liver receives little LDL cholesterol.

Distinctions that matter

· Blocking synthesis and clearing plasma LDL are two different effects, and the second depends entirely on receptor genes that can respond.

Common misconceptions

· Assuming a high plasma cholesterol suppresses hepatic synthesis. The hepatocyte senses its own endoplasmic reticulum sterol, which stays low without receptor uptake.

Where it is taught: Section C03, paragraph on how lowering liver cholesterol lowers plasma LDL, and the stop and predict box of the CV-13 Cardiovascular Biochemistry and Nutrition notes.

NPLEX-P1-CARD-13-01-C04-BD1

Brain dump 1 · foundational retrieval: Cardiac fuels and the carnitine shuttle

A competent response contains

· Fatty acids enter through the carnitine shuttle and then beta-oxidation, and their share rises in fasting, the resting adult state, and diabetes.

· Glucose enters through glycolysis and then pyruvate dehydrogenase, and its share rises with high insulin after a carbohydrate meal, in ischemia where glycolysis is anaerobic, and in the fetal heart.

· Lactate is converted by lactate dehydrogenase to pyruvate, and its share rises in exercise.

· Ketone bodies enter through succinyl-CoA:3-ketoacid CoA transferase to acetyl-CoA, and their share rises in prolonged fasting and diabetic ketoacidosis.

· At rest about 60 to 90% of cardiac ATP comes from fatty acids, which arrive bound to albumin or are released by cardiac lipoprotein lipase, enter mainly through CD36, and are activated to fatty acyl-CoA in the cytosol.

· Carnitine shuttle: CPT-I on the outer membrane forms acylcarnitine, a translocase exchanges it into the matrix for carnitine, and CPT-II re-forms acyl-CoA for beta-oxidation; short- and medium-chain fatty acids need no carnitine.

· Malonyl-CoA inhibits CPT-I; acetyl-CoA carboxylase makes it, insulin and high glucose raise it, and AMPK inhibits the carboxylase and so lowers it.

· The liver makes ketone bodies but cannot use them because it lacks the transferase that heart, muscle, and brain have.

· Creatine kinase carries high-energy phosphate from mitochondria to myofibrils as phosphocreatine, which also buffers ATP for the first seconds of a shortfall.

· The heart stores only seconds of ATP, mitochondria fill about a third of a ventricular myocyte, O2 extraction is already high at rest, so greater demand is met mainly by raising coronary flow.

Distinctions that matter

· Carnitine or CPT-II defects can cause cardiomyopathy, while short- and medium-chain fatty acids bypass the shuttle entirely.

· Malonyl-CoA is a signal at the heart rather than a building block: it reports carbohydrate abundance by closing the door to fatty acid entry.

Common misconceptions

· Saying the heart raises O2 extraction to meet demand. Extraction is already high at rest, so flow has to rise.

· Assuming all fatty acids need carnitine. Only long-chain acyl groups do.

Where it is taught: Section C04, opening paragraph, cardiac fuels table, and the carnitine shuttle paragraph of the CV-13 Cardiovascular Biochemistry and Nutrition notes.

NPLEX-P1-CARD-13-01-C04-BD2

Brain dump 2 · integration and application: Work out the oxygen cost of each fuel

A competent response contains

· Glucose consumes 6 O2, which is 12 oxygen atoms, and yields about 31 ATP, so about 2.6 ATP per oxygen atom.

· Palmitate gives 8 acetyl-CoA worth 80 ATP, 7 NADH worth 17.5, and 7 FADH2 worth 10.5, totaling 108, minus 2 for activation, so 106 ATP.

· Palmitate consumes 23 O2, which is 46 oxygen atoms, so about 2.3 ATP per oxygen atom.

· Palmitate yields over three times as much ATP per molecule, while glucose yields roughly 10% more ATP for the same O2.

· When O2 is limited but not absent, shifting toward glucose oxidation gets more work from the same coronary flow, which is why fuel selection matters in ischemia.

· Glucose and insulin raise malonyl-CoA through acetyl-CoA carboxylase, and malonyl-CoA inhibits CPT-I, so long-chain fatty acyl groups cannot enter the matrix.

· In the other direction, active fatty acid oxidation raises acetyl-CoA and NADH, which activate pyruvate dehydrogenase kinase and so inactivate pyruvate dehydrogenase, and raises citrate, which inhibits phosphofructokinase-1; this is the glucose-fatty acid cycle.

Distinctions that matter

· ATP per molecule and ATP per oxygen atom give opposite answers, and only the second one matters when flow is the limiting factor.

· Fatty acids suppress glucose use at two points, one in glycolysis through citrate and one at the gateway to the TCA cycle through pyruvate dehydrogenase.

Common misconceptions

· Concluding that fatty acids are always the better cardiac fuel because they yield more ATP per molecule. Per oxygen atom they yield less.

· Reporting absolute ATP totals as fixed. They depend on the P/O ratios used, and older texts give higher numbers without changing the comparison.

Where it is taught: Section C04, worked example on oxygen cost and the carnitine shuttle paragraph of the CV-13 Cardiovascular Biochemistry and Nutrition notes.

NPLEX-P1-CARD-13-01-C04-RT01

Retrieval exercise: Rebuild energy failure in ischemia

A competent response contains

· Step 1: flow stops and myocardial O2 is exhausted within seconds.

· Step 2: oxidative phosphorylation, the TCA cycle, and beta-oxidation halt.

· Step 3: phosphocreatine falls first, then ATP, while AMP and phosphate rise.

· Step 4: glycogenolysis and anaerobic glycolysis make lactate, yielding only 2 ATP per glucose and 3 per glucose unit from glycogen, with lactate dehydrogenase regenerating the NAD+ glycolysis needs.

· Step 5: lactate and H+ accumulate, and acidosis inhibits phosphofructokinase-1, which limits the one remaining ATP source.

· Step 6: phosphate and H+ weaken cross-bridges and contraction fails within about a minute, which is reversible and spares the remaining ATP.

· Step 7: pump failure causes K+ loss and Ca2+ overload.

· Step 8: injury becomes irreversible after about 20 to 40 minutes.

· Ion sequence: the Na+/K+-ATPase slows, K+ leaks out through ATP-sensitive channels, intracellular Na+ rises, the Na+/Ca2+ exchanger extrudes less Ca2+ while Na+/H+ exchange adds more Na+, cytosolic Ca2+ rises, and SERCA failure impairs relaxation.

· Troponin is released once the membrane is irreversibly damaged, and lactate appearing in coronary sinus blood marks ischemia because the normal heart takes lactate up.

Distinctions that matter

· Losing contraction early is protective rather than part of the damage, because it conserves what ATP is left.

· Raised local extracellular K+ depolarizes ischemic cells, which links energy failure to conduction slowing and arrhythmia.

Common misconceptions

· Expecting anaerobic glycolysis to sustain the heart. Two ATP per glucose is far short of demand, and its own acid product shuts it down.

· Treating contractile failure as the moment of cell death. Injury is still reversible for roughly 20 to 40 minutes.

Where it is taught: Section C04, flow diagram from coronary occlusion to energy failure and the paragraph that follows of the CV-13 Cardiovascular Biochemistry and Nutrition notes.

NPLEX-P1-CARD-13-01-C04-PR01

Prediction exercise: Predict what happens when the carnitine shuttle fails

A competent response contains

· Long-chain fatty acyl-CoA accumulates in the cytosol, because CPT-II cannot re-form acyl-CoA in the matrix and the shuttle backs up.

· Beta-oxidation falls, because its substrate never reaches the matrix.

· Mitochondrial acetyl-CoA and NADH fall, so pyruvate dehydrogenase kinase is less active and pyruvate dehydrogenase activity rises, leaving the myocyte more dependent on glucose oxidation.

· ATP production falls during the fast, when fatty acids should be supplying 60 to 90% of resting cardiac ATP, and the chronically energy-starved ventricle develops cardiomyopathy.

· Medium-chain fatty acids are oxidized normally, because short- and medium-chain fatty acids enter the matrix without carnitine.

· Glucose raises malonyl-CoA through acetyl-CoA carboxylase, which inhibits CPT-I and shifts the heart further toward glucose oxidation, so glucose both supplies fuel and reduces the futile attempt at fatty acid entry.

Distinctions that matter

· A block at CPT-II leaves the cytosolic activation steps intact, so the problem is transport into the matrix rather than failure to activate the fatty acid.

· Carnitine and CPT-II defects both cause cardiomyopathy, and both spare medium-chain fatty acid oxidation.

Common misconceptions

· Expecting ketone bodies to rescue the heart here. Hepatic ketogenesis also depends on fatty acid oxidation, so the alternative fuel is limited at the same time.

· Assuming all fatty acids are blocked. Chain length decides whether the shuttle is needed.

Where it is taught: Section C04, carnitine shuttle paragraph and the cardiac fuels table of the CV-13 Cardiovascular Biochemistry and Nutrition notes.

NPLEX-P1-CARD-13-01-C05-BD1

Brain dump 1 · foundational retrieval: Thiamine pyrophosphate and the enzymes that need it

A competent response contains

· Thiamine is absorbed mainly in the jejunum and converted by thiamine pyrophosphokinase to thiamine pyrophosphate; stores are small and turnover is fast, so deficiency can appear within weeks.

· The thiazolium carbanion of the coenzyme attacks the carbonyl carbon of an alpha-keto acid, releasing CO_2 and carrying the acyl fragment, which is the chemistry both oxidative decarboxylation and transketolase need.

· Pyruvate dehydrogenase E1 converts pyruvate to acetyl-CoA, and deficiency lets pyruvate accumulate and become lactate.

· Alpha-ketoglutarate dehydrogenase E1 converts alpha-ketoglutarate to succinyl-CoA, and deficiency slows the TCA cycle, cutting NADH and ATP from every fuel.

· Branched-chain alpha-ketoacid dehydrogenase catabolizes leucine, isoleucine, and valine, and deficiency lets those keto acids accumulate, which is minor in acquired deficiency.

· Transketolase transfers two carbons in the pentose phosphate pathway, and deficiency lowers red cell activity, which is used as a test.

· The dehydrogenase complexes also need lipoic acid, CoA from B5, FAD from B2, and NAD+ from B3.

· The test is red cell transketolase activity: a large rise when thiamine pyrophosphate is added to the sample indicates deficiency.

· Tissues with high aerobic glucose use are hit first, which is why the nervous system and the heart show the earliest effects.

Distinctions that matter

· Two of the four enzymes sit on the main fuel route and two do not, which is why lactic acidosis and low red cell transketolase appear together.

· Thiamine pyrophosphate is one of five cofactors the dehydrogenase complexes need, so the complexes can fail for reasons other than thiamine.

Common misconceptions

· Naming thiamine itself as the coenzyme. Thiamine pyrophosphokinase has to add the pyrophosphate first.

· Expecting slow onset because the vitamin is water-soluble. Stores are small and turnover fast, so weeks are enough.

Where it is taught: Section C05, opening paragraph and the table of thiamine pyrophosphate enzymes of the CV-13 Cardiovascular Biochemistry and Nutrition notes.

NPLEX-P1-CARD-13-01-C05-BD2

Brain dump 2 · integration and application: Explain every finding in wet beriberi

A competent response contains

· Mechanism in order: low thiamine pyrophosphate blocks pyruvate and alpha-ketoglutarate dehydrogenase; aerobic ATP falls while pyruvate and lactate accumulate; metabolic vasodilation lowers systemic vascular resistance; venous return and cardiac output rise; sympathetic and renin-angiotensin-aldosterone activation retain Na+ and water; the ATP-starved myocardium dilates and fails.

· Warm extremities and a bounding pulse with a wide pulse pressure come from low resistance with a high stroke volume; tachycardia comes from sympathetic drive; raised jugular venous pressure, edema, and pulmonary congestion come from salt and water retention; the enlarged heart on radiograph comes from dilation of the failing myocardium; the raised lactate comes from the enzyme block itself.

· Lactate is high with normal oxygenation because pyruvate dehydrogenase cannot move pyruvate into the TCA cycle, so pyruvate accumulates and is reduced to lactate; the problem is fuel entry, not O2 delivery.

· Shoshin beriberi is the fulminant form, with severe lactic acidosis and circulatory collapse.

· Settings: alcohol use disorder, which impairs intake, absorption, storage, and conversion to the coenzyme; polished rice diets; persistent vomiting or bariatric surgery; and long diuretic use.

· A glucose load increases flux through thiamine pyrophosphate enzymes, so it can unmask marginal stores.

· Separating the other high-output states: severe anemia shows a low hemoglobin, thyrotoxicosis shows a low TSH with a high free T4 or T3, and an arteriovenous fistula has a continuous bruit with a heart rate that falls on compression.

· Ordinary low-output failure gives cool extremities and a narrow pulse pressure.

Distinctions that matter

· Low resistance and a failing myocardium come from the same metabolic block, so this is one disease rather than two coexisting problems.

· Dry beriberi is the peripheral neuropathy, and Wernicke encephalopathy with Korsakoff syndrome are the central forms of the same deficiency.

Common misconceptions

· Reading a raised lactate as proof of poor oxygen delivery. Here the cell has O2 but cannot feed fuel into the TCA cycle to use it.

· Expecting cool extremities in heart failure. High-output failure runs warm with a wide pulse pressure.

Where it is taught: Section C05, wet beriberi flow diagram, the paragraph on findings, and the stop and predict box of the CV-13 Cardiovascular Biochemistry and Nutrition notes.

NPLEX-P1-CARD-13-01-C05-CA01

Clinical application: Breathlessness and warm hands in a malnourished man

A competent response contains

· Recognize: thiamine deficiency causing wet beriberi, a high-output heart failure; the warm extremities and the wide pulse pressure of 134/48 mm Hg with a bounding pulse show output is high rather than low.

· Explain: pyruvate dehydrogenase and alpha-ketoglutarate dehydrogenase lose thiamine pyrophosphate, so aerobic ATP falls and pyruvate and lactate accumulate; the accumulating metabolites drive peripheral vasodilation that lowers systemic vascular resistance and raises venous return and output, while the same ATP shortage leaves the myocardium unable to sustain that work, so it dilates and fails, and sympathetic and renin-angiotensin-aldosterone activation retains salt and water.

· Differentiate: severe anemia is excluded by a hemoglobin of 13.1 g/dL, thyrotoxicosis by a normal TSH, an arteriovenous fistula by the absence of a continuous bruit and a heart rate that would fall on compression, and ordinary low-output failure by his warm extremities and wide pulse pressure rather than cool extremities and a narrow pulse pressure.

· Investigate: red cell transketolase activity measured before and after adding thiamine pyrophosphate, where a large rise in activity on adding the coenzyme indicates deficiency; the lactate itself is the test that reflects the blocked entry of pyruvate into the TCA cycle.

· Decide: lactate is raised despite 97% saturation because the block is at fuel entry rather than O2 delivery, and with a lactate of 6.2 mmol/L and a low diastolic pressure the fulminant form to watch for is Shoshin beriberi with severe lactic acidosis and circulatory collapse.

· Justify: the thiazolium carbanion of thiamine pyrophosphate attacks the carbonyl carbon of an alpha-keto acid and releases CO2, which is what makes oxidative decarboxylation possible; without it pyruvate

accumulates behind the block and is reduced to lactate, and the nervous system and heart fail first because they depend most on high aerobic glucose use and have the least metabolic slack.

Distinctions that matter

· High-output failure and low-output failure are separated at the bedside by skin temperature and pulse pressure, before any laboratory test.

· His alcohol history explains the deficiency, but the diagnosis rests on the pattern of findings and the enzyme block, not on the history alone.

Common misconceptions

· Looking for a cause of poor oxygen delivery to explain the lactate. Oxygenation is normal and the defect is the coenzyme.

· Attributing the edema and raised jugular venous pressure to a failing pump alone. Sympathetic and renin-angiotensin-aldosterone driven salt and water retention is part of the mechanism.

Where it is taught: Section C05, wet beriberi flow diagram, the paragraph on findings, and the stop and predict box of the CV-13 Cardiovascular Biochemistry and Nutrition notes.

NPLEX-P1-CARD-13-01-C05-Q03

Board-style question: A high lactate with normal oxygenation

Correct answer: D. Pyruvate dehydrogenase and alpha-ketoglutarate dehydrogenase

Why it is right: Transketolase and both of these dehydrogenase complexes use thiamine pyrophosphate, so a large rise in red cell transketolase activity on adding the coenzyme shows thiamine deficiency; the block at pyruvate dehydrogenase raises lactate and the block at alpha-ketoglutarate dehydrogenase cuts aerobic ATP, which together give the warm, bounding, high-output picture of wet beriberi.

A is wrong: The carnitine palmitoyltransferases need no thiamine, and their failure causes cardiomyopathy during fasting rather than lactic acidosis with a transketolase response.

B is wrong: Methionine synthase and cystathionine beta-synthase use B12 and B6 and would raise homocysteine, not lactate.

C is wrong: Lysyl oxidase and prolyl hydroxylase need copper and ascorbate and would weaken collagen and elastin, giving bleeding or fragile vessels rather than this circulatory picture.

A competent response contains

· Persistent vomiting after bariatric surgery is one of the listed settings for thiamine deficiency, and a sharp rise in red cell transketolase activity when thiamine pyrophosphate is added confirms it.

· The two dehydrogenase complexes that fail first are pyruvate dehydrogenase, which lets pyruvate accumulate and become lactate, and alpha-ketoglutarate dehydrogenase, which slows the TCA cycle.

Distinctions that matter

· The transketolase test identifies the missing coenzyme, while the clinical picture of warm extremities and a wide pulse pressure identifies which organ system is failing.

Common misconceptions

· Taking the normal oxygen saturation as evidence against a metabolic cause of the lactate. Here the cell cannot use the O2 it has.

Where it is taught: Section C05 of the CV-13 Cardiovascular Biochemistry and Nutrition notes.

NPLEX-P1-CARD-13-01-C06-BD1

Brain dump 1 · foundational retrieval: Draw the methionine cycle and its two exits

A competent response contains

· Cycle in order: methionine is activated by ATP to S-adenosylmethionine, which donates a methyl group and becomes S-adenosylhomocysteine, and hydrolysis of that yields homocysteine.

· Remethylation exit: methionine synthase uses B12 as methylcobalamin and N5-methyltetrahydrofolate as its methyl donor, regenerating tetrahydrofolate; MTHFR makes that methyl donor; the liver also remethylates through betaine-homocysteine methyltransferase.

· Transsulfuration exit: cystathionine beta-synthase commits homocysteine to cystathionine and cystathionine gamma-lyase completes the step to cysteine, both using pyridoxal phosphate from B6.

· Transsulfuration is the only exit that removes homocysteine permanently; remethylation returns it to methionine.

· Homocysteine leaves only through these enzymes, so loss of B12, folate, B6, or the enzymes themselves makes it accumulate.

· The methylfolate trap: in B12 deficiency, folate is stuck as N5-methyltetrahydrofolate because methionine synthase cannot use it, so both deficiencies cause megaloblastic anemia.

· Methylmalonic acid separates them: it is raised only in B12 deficiency, because B12 is also the cofactor for methylmalonyl-CoA mutase.

Distinctions that matter

· Folate and B12 act on the same step from different sides, one as the methyl donor and one as the cofactor, which is why their blood pictures look alike.

· B6 serves the exit that disposes of homocysteine, so B6 deficiency raises homocysteine without a megaloblastic anemia.

Common misconceptions

· Treating remethylation as disposal. It recycles homocysteine back to methionine, so only transsulfuration clears it.

· Using megaloblastic anemia to tell folate from B12 deficiency. Methylmalonic acid is the value that separates them.

Where it is taught: Section C06, methionine cycle diagram and the paragraph on remethylation and transsulfuration of the CV-13 Cardiovascular Biochemistry and Nutrition notes.

NPLEX-P1-CARD-13-01-C06-BD2

Brain dump 2 · integration and application: Two ways micronutrients let a vessel fail

A competent response contains

· Homocysteine oxidizes readily and generates reactive oxygen species that injure endothelium; it reduces nitric oxide; it promotes smooth muscle proliferation; and it favors thrombosis, for example by raising tissue factor and impairing the thrombomodulin and protein C pathway.

· Normal fasting total homocysteine is about 5 to 15 micromol per litre, and a mild elevation is best read as a risk marker, because whether it causes atherosclerosis or only accompanies it is unsettled.

· The infant has Menkes disease: copper-dependent lysyl oxidase has failed, so elastin and collagen cannot be cross-linked, the aldehyde-derived covalent cross-links including desmosine in elastin are missing, and the aorta and large arteries lose strength and recoil and become tortuous and aneurysmal.

· The older man has scurvy: prolyl and lysyl hydroxylases have failed because ascorbate is needed to keep their iron as Fe2+, so hydroxyproline cannot stabilize the collagen triple helix at 37 degrees C (98.6 degrees F) and hydroxylysine cannot provide glycosylation and cross-link sites; weak collagen in capillary basement membranes and perivascular tissue gives the bleeding.

· Location: the hydroxylases act inside the cell in the rough endoplasmic reticulum, while lysyl oxidase acts extracellularly after secretion.

· Menkes disease is X-linked recessive from ATP7A, so copper cannot leave the enterocyte and serum copper and ceruloplasmin are low; acquired copper deficiency follows bariatric surgery or excess zinc; Wilson disease from ATP7B is the opposite, with copper retained in liver and brain.

· Other scurvy findings are perifollicular hemorrhages, bleeding gums, hemarthrosis, subperiosteal bleeding, poor wound healing, and corkscrew hairs, and stores last about 1 to 3 months.

Distinctions that matter

· Vitamin C is needed to build the helix and lay down cross-link sites, while copper is needed to form the cross-links themselves, which is why one causes bleeding and the other causes fragile tortuous arteries.

· Menkes and Wilson disease are both copper transport defects in the same family of proteins with opposite consequences.

Common misconceptions

· Explaining the bleeding of scurvy as a platelet or clotting problem. The collagen supporting small vessels is weak.

· Treating a mildly raised homocysteine as established cause of atherosclerosis. Homocystinuria shows the extreme, but a mild rise is a marker.

Where it is taught: Section C06, paragraph on vascular injury and the final two paragraphs on vitamin C and copper of the CV-13 Cardiovascular Biochemistry and Nutrition notes.

NPLEX-P1-CARD-13-01-C06-CP01

Comparison table: Sorting out a raised homocysteine

A competent response contains

· Folate deficiency: remethylation is affected, methylmalonic acid is normal, and megaloblastic anemia is the other finding.

· Vitamin B12 deficiency: remethylation is affected, methylmalonic acid is raised, and there is megaloblastic anemia with neurologic signs.

· Vitamin B6 deficiency: transsulfuration is affected, methylmalonic acid is normal, and there is sideroblastic anemia with neuropathy, classically with isoniazid.

· Cystathionine beta-synthase deficiency, homocystinuria: transsulfuration is affected, methylmalonic acid is normal, inheritance is autosomal recessive, and homocysteine is often above 100 micromol per litre with raised methionine, downward lens dislocation, a marfanoid habitus, and early thrombosis.

· MTHFR C677T variant: methyl donor formation is affected, methylmalonic acid is normal, and the rise is mild and appears mainly with low folate.

· Chronic kidney disease: clearance is reduced, methylmalonic acid may be raised, and the clue is a low glomerular filtration rate.

· Normal fasting total homocysteine is about 5 to 15 micromol per litre.

Distinctions that matter

· Methylmalonic acid is the single value that separates B12 deficiency from folate deficiency, because B12 also serves methylmalonyl-CoA mutase.

· A very high homocysteine with a raised methionine points to the transsulfuration enzyme itself, since the remethylation causes leave methionine low or normal.

Common misconceptions

· Expecting a megaloblastic anemia with every cause of raised homocysteine. B6 deficiency and cystathionine beta-synthase deficiency affect the other exit.

· Reading the MTHFR variant as a strong independent cause. It gives a mild rise, mainly when folate is low.

Where it is taught: Section C06, table of causes of raised homocysteine of the CV-13 Cardiovascular Biochemistry and Nutrition notes.

NPLEX-P1-CARD-13-01-C07-BD1

Brain dump 1 · foundational retrieval: Map each electrolyte onto the action potential and the tracing

A competent response contains

· Action potential to ECG: phase 4 rest near -90 mV is set by K+, phase 0 is the Na+ upstroke and gives the QRS, phase 2 is the Ca2+ plateau and corresponds to the ST segment, and phase 3 is K+ repolarization and gives the T wave.

· Hyperkalemia: phase 3 is faster and the resting membrane depolarizes with Na+ channel inactivation, so the tracing shows a peaked T wave, then a flat P, a long PR, a wide QRS, and finally a sine wave, with ventricular fibrillation and asystole as the rhythm risks.

· Hypokalemia: phase 3 is slower, so T waves flatten or invert, the ST segment sags, U waves appear with a long QU interval, and the risks are torsades and other tachyarrhythmias with increased glycoside toxicity.

· Hypocalcemia: phase 2 is longer, so the ST segment and QT are long with a normal T wave, and torsades is possible though uncommon, alongside tetany.

· Hypercalcemia: phase 2 is shorter, so the ST segment and QT are short, with bradyarrhythmia and AV block at high levels.

· Hypomagnesemia: repolarization is long, mostly through low K+ and Ca2+, giving a long QT often with hypokalemic features and a risk of torsades de pointes.

· Ranges: K+ 3.5 to 5.0 mEq/L, total calcium 8.5 to 10.5 mg/dL with ionized calcium about 4.5 to 5.3 mg/dL, and Mg2+ about 1.7 to 2.2 mg/dL.

· Add 0.8 mg/dL to total calcium for each 1 g/dL that albumin is below 4.0 g/dL; hemolysis falsely raises a potassium result; and plasma magnesium can be normal despite real cellular depletion.

· Kidney failure often combines hyperkalemia and hypocalcemia, giving a long ST segment followed by a peaked T wave.

Distinctions that matter

· Potassium acts on phase 3 and the T wave first, while calcium acts on phase 2 and the length of the ST segment, so the two disorders change different parts of the tracing.

· Hypokalemia and hyperkalemia both act on K+ conductance but in opposite directions, which is why one slows phase 3 and the other speeds it.

Common misconceptions

· Expecting a low extracellular K+ to speed repolarization because K+ efflux is repolarizing. Low K+ reduces IK1 and IKr conductance, so phase 3 slows.

· Treating a normal tracing as reassurance in hyperkalemia. The level at which each change appears varies widely and a normal ECG does not exclude danger.

Where it is taught: Section C07, opening paragraph, the action potential diagram, and the electrolyte table of the CV-13 Cardiovascular Biochemistry and Nutrition notes.

NPLEX-P1-CARD-13-01-C07-BD2

Brain dump 2 · integration and application: Why one potassium goes high and another will not come up

A competent response contains

· The first change is a tall, narrow, peaked T wave, because higher extracellular K+ increases IKr and inward rectifier conductance and speeds phase 3; repolarizing currents respond at moderate levels before anything else.

· QRS widening needs enough resting depolarization to inactivate a large fraction of fast Na+ channels, which slows phase 0 and conduction; the order is a flattening P wave, then a lengthening PR, then a widening QRS, then a sine wave pattern that can lead to ventricular fibrillation or asystole.

· In the woman with refractory hypokalemia, measure magnesium: Mg2+ normally blocks K+ exit through renal ROMK channels, so hypomagnesemia causes continued urinary K+ loss and the hypokalemia persists until magnesium is restored.

· The two other consequences of magnesium deficiency given in the notes are impaired parathyroid hormone release and action, which causes hypocalcemia, and a predisposition to torsades de pointes.

· Torsades arises when a long QT allows early afterdepolarizations, driven by reactivation of L-type Ca2+ channels, to reach threshold, often after a short-long-short beat sequence; the QRS axis appears to twist around the baseline.

· Low K+ increases cardiac glycoside toxicity because K+ competes with these drugs at the Na+/K+-ATPase.

· Hypomagnesemia also promotes torsades by lowering K+ and Ca2+ and by removing its own limit on inward Ca2+ current, and its causes include alcohol use disorder, diarrhea, loop and thiazide diuretics, and long-term proton pump inhibitors.

Distinctions that matter

· Hyperkalemia changes repolarization first and depolarization later, which is why the T wave is the early sign and the wide QRS is the late one.

· A potassium that will not correct is a magnesium problem at the renal tubule, not a dosing problem.

Common misconceptions

· Reading a wide QRS as the first sign of hyperkalemia. Peaked T waves come first, at lower levels.

· Assuming a normal plasma magnesium rules out depletion. Plasma can be normal while cells are depleted.

Where it is taught: Section C07, potassium paragraphs, the calcium and magnesium paragraph, and the stop and predict box of the CV-13 Cardiovascular Biochemistry and Nutrition notes.

NPLEX-P1-CARD-13-01-C07-PR01

Prediction exercise: Predict the tracing in kidney failure

A competent response contains

· Phase 3 is faster and the T wave becomes tall, narrow, and peaked, because the raised extracellular K+ increases IKr and inward rectifier conductance.

· Phase 2 is longer and the ST segment lengthens, because slower Ca2+-dependent inactivation of L-type channels at a low ionized calcium prolongs the plateau.

· The QT interval is long, and the hypocalcemia is driving it through the lengthened phase 2, while the T wave itself stays normal in shape from the calcium effect and peaked from the potassium effect; with an albumin of 4.0 g/dL no correction to total calcium is needed, so the value of 6.8 mg/dL is a true hypocalcemia.

· The resting membrane potential becomes less negative, and a growing fraction of fast Na+ channels is inactivated, because the resting potential lies near the K+ equilibrium potential and a raised extracellular K+ shifts it upward.

· As K+ rises further the P wave flattens, the PR interval lengthens, and the QRS widens, because slowed phase 0 slows conduction through atrium, node, and ventricle.

· The tracing becomes a sine wave pattern, and the rhythms to expect are ventricular fibrillation and asystole.

· This combination is the classic kidney failure tracing: a long ST segment followed by a peaked T wave.

Distinctions that matter

· The two abnormalities act on different parts of the tracing, so the ST segment reports the calcium and the T wave reports the potassium.

· Albumin correction only applies when albumin is below 4.0 g/dL, so here the measured total calcium stands as reported.

Common misconceptions

· Expecting a long QT to come from the hyperkalemia. Raised K+ speeds phase 3 and shortens repolarization; the long segment here is the calcium plateau.

· Assuming the level of potassium predicts the finding reliably. Thresholds vary widely between patients.

Where it is taught: Section C07, potassium paragraphs, calcium and magnesium paragraph, and the electrolyte table with its caption of the CV-13 Cardiovascular Biochemistry and Nutrition notes.

NPLEX-P1-CARD-13-01-C07-Q04

Board-style question: A potassium that will not correct

Correct answer: B. Magnesium depletion has removed its normal block of renal ROMK channels, so potassium continues to be lost in the urine

Why it is right: Magnesium normally blocks K+ exit through renal ROMK channels, so while magnesium is depleted the kidney keeps losing potassium and replacement cannot raise the plasma level; her thiazide and long-term proton pump inhibitor are both listed causes of magnesium loss, and low magnesium also explains the long corrected QT and the polymorphic ventricular tachycardia.

A is wrong: Impaired parathyroid hormone release does follow severe magnesium deficiency and lengthens phase 2 through hypocalcemia, but it explains a long ST segment rather than why the potassium itself will not rise.

C is wrong: Her extracellular potassium is low, not raised, and a raised potassium would speed phase 3 and peak the T waves instead of flattening them and adding U waves.

D is wrong: Alkalosis increasing albumin binding of calcium causes tetany with a normal total calcium; it does not cause renal potassium loss.

A competent response contains

· Mg2+ normally blocks K+ exit through renal ROMK channels, so hypomagnesemia causes ongoing urinary K+ loss and a hypokalemia that persists until magnesium is restored.

· Thiazide diuretics and long-term proton pump inhibitors are both listed causes of magnesium loss, and low magnesium also promotes torsades through low K+ and Ca2+ and by removing its limit on inward Ca2+ current.

Distinctions that matter

· Flat T waves with U waves and a long corrected QT is the hypokalemic pattern, and the failure to correct points past the potassium to the magnesium.

Common misconceptions

· Treating a persistently low potassium as inadequate replacement. The renal leak continues while magnesium is low.

Where it is taught: Section C07, calcium and magnesium paragraph of the CV-13 Cardiovascular Biochemistry and Nutrition notes.